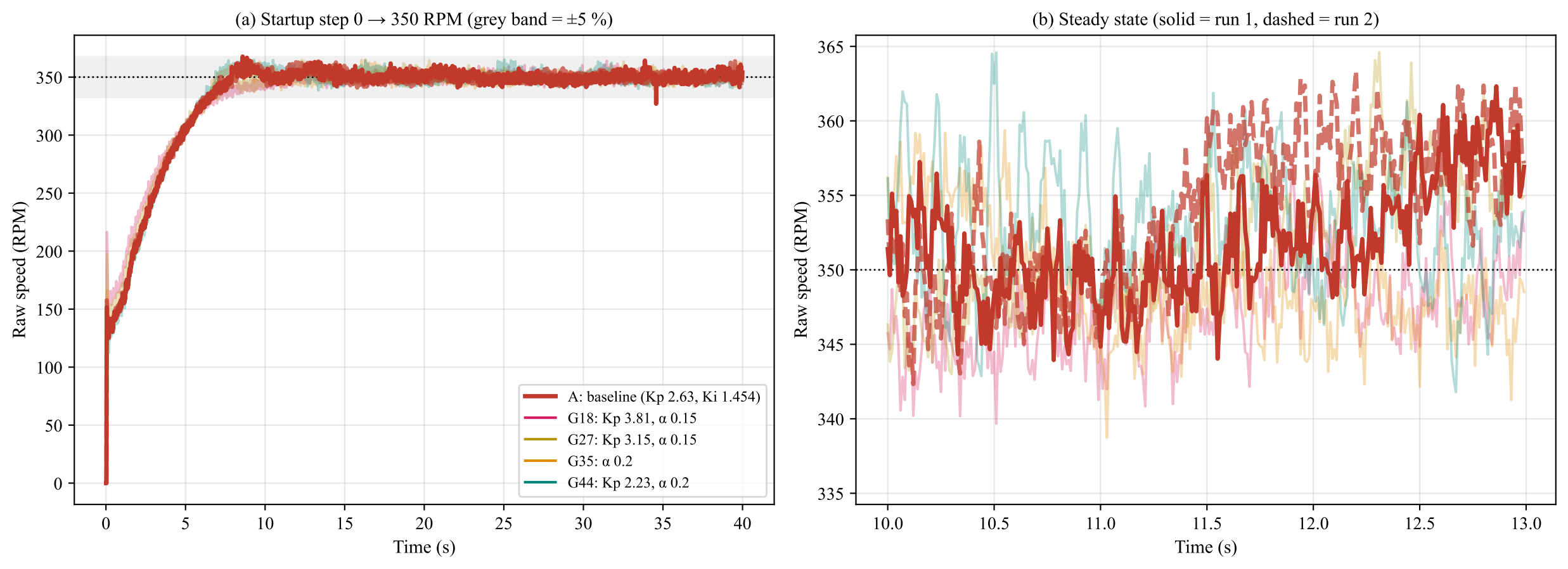

(a) Startup step 0 → 350 RPM (grey band = ±5 %)
Raw speed (RPM)
Time (s)
A: baseline (Kp 2.63, Ki 1.454)
G18: Kp 3.81, α 0.15
G27: Kp 3.15, α 0.15
G35: α 0.2
G44: Kp 2.23, α 0.2
(b) Steady state (solid = run 1, dashed = run 2)
Raw speed (RPM)
Time (s)